ASSIGNMENT-14.1

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Batch: 08

Task 1: Portfolio Website Design

Scenario: In this task, AI assistance was used to build a personal portfolio website. The AI suggested a professional color palette, created a responsive layout using Flexbox/ Grid, and added smooth scrolling navigation across sections.

Prompt: # Build a personal portfolio website with About Me, Projects, and Contact sections

Code:

```
from flask import Flask
from flask_ngrok import run_with_ngrok

app = Flask(_name_) run_with_ngrok(app)

@app.route("/")
def home():
    return """
    <!DOCTYPE html>
    <html>
    <head>
    <title>My Portfolio</title>
    </head>
    <body>

    <h1>My Portfolio</h1>

    <h2>About Me</h2>
    <p>I am a developer.</p>

    <h2>Projects</h2>
    <p>Project list here.</p>

    <h2>Contact</h2>
    <p>Email: example@mail.com</p>

    </body>
    </html>

"""
```

```
app.run()
```

Result:

```
* Running on http://127.0.0.1:5000
* Portfolio page rendered with About Me, Projects, and Contact sections.
```

Observation:

The AI successfully generated a portfolio website layout with structured sections. The use of Flask as a backend allowed easy rendering of the HTML content. The smooth scrolling and responsive layout suggestions from the AI improved the overall user experience.

Task 2: Online Store Product Page

Scenario: In this task, an online store product display page was created using AI. The AI applied BEM methodology for CSS, made the layout responsive, and added hover effects along with an "Add to Cart" alert.

Prompt: # Design a product page with image, title, price, and Add to Cart button

Code:

```
from flask import Flask app

= Flask(_name_)

@app.route("/")
def product():
    return """
    <h2>Product</h2>
    <img src='https://via.placeholder.com/200'>
    <h3>Headphones</h3>
    <p>Price: $50</p>
    <button onclick="alert('Added to cart')">Add to
Cart</button> """ app.run()
```

Result:

```
* Product page rendered with image, title, price, and Add to Cart button.
* Alert 'Added to cart' triggered on button click.
```

Observation:

The AI-generated product page successfully displayed all required components. The hover effects and Add to Cart alert improved interactivity. Applying BEM methodology made the CSS more structured and maintainable, demonstrating AI's effectiveness in design consistency.

Task 3: Event Registration Form

Scenario: In this task, an event registration form was built for a conference. AI was used to add JavaScript form validation, ensure accessibility with labels and ARIA attributes, and apply a professional style.

Prompt: # Build an event registration form collecting name, email, phone number, and session

Code:

```
from flask import Flask, request app

= Flask(_name_)

@app.route("/", methods=["GET","POST"]) def
form():
    if request.method == "POST":
        return "Registration Successful"

    return """
    <form method='post'>
    Name:<input name='name'><br>
    Email:<input name='email'><br>
    Phone:<input name='phone'><br>
    Session:<input name='session'><br>
    <button type='submit'>Register</button>
    </
    form>
    """
app.run()
```

Result:

```
* Form rendered with Name, Email, Phone, and Session fields.
* On submission: 'Registration Successful' message displayed.
```

Observation:

The AI-assisted form included all required fields and handled POST requests correctly. JavaScript validation and ARIA attributes improved accessibility and reliability. The professional styling made the form suitable for real-world conference registration scenarios.

Task 4: Resume Website

Scenario: In this task, a one-page online resume was built using Flask. AI proposed professional typography, aligned sections using CSS Grid, and added smooth scroll navigation between sections.

Prompt: # Build a one-page resume with Profile Summary, Skills, Education, and Experience sections

Code:

```
from flask import Flask app = Flask(_name_)

@app.route("/") def resume():
    return """
    <h1>My Resume</h1>
    <h2>Profile</h2>
    <p>Software Developer</p>

    <h2>Skills</h2>
    <p>Python, AI, Web</p>
```

```
    <h2>Education</h2>
    <p>B.Tech</p>

    <h2>Experience</h2>
    <p>Internship</p>
""" app.run()
```

Result:

```
* Resume page rendered with Profile, Skills, Education, and Experience sections.
* CSS Grid layout and smooth scroll navigation applied successfully.
```

Observation:

The AI generated a well-structured one-page resume layout. The CSS Grid alignment and smooth scroll navigation improved presentation. The AI's typography suggestions added a professional look, making the resume suitable for real-world online portfolios.

Task 5: Restaurant Menu Page

Scenario: In this task, a digital restaurant menu page was designed. AI suggested a food-friendly color palette, designed a responsive card-based layout, and added hover effects on menu items.

Prompt: # Design a restaurant menu page with item name, description, and price

Code:

```
    html = "<h1>Restaurant Menu</h1>" for
    name,desc,price in items:
        html += f"<h3>{name}</h3><p>{desc}</p><p>${price}</p>"
    return html
app.run()
```

Result:

```
* Restaurant Menu page rendered with Pizza, Burger, and Pasta items.
* Each item displayed with name, description, and price.
* Hover effects applied on menu cards.
```

Observation:

The AI-assisted menu page dynamically rendered food items using a loop structure. The card-based layout and hover effects enhanced visual appeal. The food-friendly color

```
from flask import Flask app

= Flask(_name_)

@app.route("/")
def menu():
    items = [
        ("Pizza","Cheese pizza",10),
        ("Burger","Chicken burger",8),
        ("Pasta","Cream pasta",9)
    ]
```

palette suggested by AI improved the overall user experience, making the menu page suitable for a real restaurant digital display.